DROIDLINK

COMPLETE USER GUIDE

DroidLink_AP

Controller selection, wiring, templates, calibration, lighting, sequences, and backups.

Complete Guide · Document revision 2
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> DroidLink_AP V2.4.0 supports the Master's on-demand device catalog. RC Actuator Controls require the Maestro or PCA9685 backend; MarcDuino remains command-only. See the Update Droid and Device Catalog Guide and RC Actuator Controls Guide.

DroidLink_AP combines AstroPixels lighting with a choice of Pololu Maestro, PCA9685, or wired Marcduino control. This guide covers hardware, wiring, installation, first-time setup, Master registration, Web Config, templates, normal operation, sequences, backups, and troubleshooting. Detailed command help is also available inside the installed firmware's Web Config interface.

Select the current **DroidLink_AP** entry in the Installer. The Firmware Changelog lists the current released version.

Choose your controller path

DroidLink_AP supports three output-controller modes. Follow only the instructions for the hardware selected during first-time setup.

- **Pololu Maestro users:** Continue through the complete Maestro wiring and Maestro Control Center sections. After installation, import the Maestro dome template. For a standard installation, ignore the advanced assignment dropdowns, scroll to the bottom, and click **Apply template**.
- **PCA9685 users:** Read **Before installation**, then skip the entire section labeled **Maestro users only**. Complete the firmware installation and Web Config steps, then import the PCA9685 dome template. PCA users do not configure device numbers in Maestro Control Center.
- **Marcduino users:** Read **Before installation**, then skip the sections labeled **Maestro users only**, **Required: install the dome output template**, and **Advanced settings**. Marcduino mode does not use the Maestro/PCA output templates or servo-calibration controls.

The normal template procedure is simple: import the correct template, leave the assignment dropdowns alone, scroll to the bottom, and click **Apply template**. The dropdowns are only for users who deliberately changed the standard wiring.

Output controls

Maestro and PCA9685 modes add **360-degree servo** as an output type. Each 360-degree servo stores Reverse, Stop, and Forward pulses within 1000-2000 us; the default Stop pulse is 1500 us. Direct commands use the numbered output shown in Web Config:

- `:DS<n>F` runs 360-degree servo output `<n>` forward.
- `:DS<n>R` runs it in reverse.
- `:DS<n>S` stops it at its saved Stop pulse.
- `:DS<n>T` toggles a normal positional-servo output `<n>` between its saved Closed and Full positions.

Custom output names do not change command targeting. Shared sequences operate the same mechanism only when the receiving droid uses the same numbered output; each droid retains its own calibration.

Before installation

Use regulated power suitable for the connected lighting and servos. Connect the grounds of the ESP32, controller boards, and servo power supply. Disconnect servo linkages while establishing safe endpoints.

Choose the controller mode that matches the installed hardware:

- **Pololu Maestro:** use the AstroPixels board's **Serial2** header labeled `G V R T`. Connect **T** to the Maestro serial input/RX and **G** to Maestro ground. Leave **V** and **R** disconnected. The firmware uses 115200 baud and defaults to Maestro device numbers 12 and 13.
- **PCA9685:** connect GPIO21 to SDA and GPIO22 to SCL. Configure the two boards as addresses `0x40` and `0x41`. Follow the AstroPixelsPlus servo wiring diagram for the standard dome layout.
- **Marcduino:** use the same AstroPixels **Serial2** header labeled `G V R T`. Connect **T** to the Marcduino serial input/RX and **G** to Marcduino ground. For two-way communication, also connect **R** to the Marcduino serial output/TX. Leave **V** disconnected.

Maestro users only: wiring and controller setup

The AstroPixelsPlus servo wiring diagram shows PCA9685 boards, but the panel and holoprojector order also applies when using Maestro controllers. Treat the first Maestro as the **Dome Panel Controller** in that diagram and the second Maestro as the **Holoprojector Controller**.

> **Important numbering rule:** Maestro channels are numbered from zero. The first physical channel on a Maestro is **channel 0**, not channel 1. Therefore dome panel **P1 plugs into the Maestro's first channel, channel 0**. P2 plugs into channel 1, P3 into channel 2, and so on. `P1` is the panel's name; it is not a Maestro channel number.

Before connecting anything

1. Turn off power to the ESP32, both Maestro controllers, AstroPixels, and the servo power supply.
2. Disconnect the servo linkages or otherwise make sure a moving panel cannot bind or damage the dome during setup.
3. Use a regulated supply suitable for the number and type of servos installed. Do not rely on USB power to operate the dome servos.
4. Connect the grounds of the ESP32, both Maestro controllers, AstroPixels, and the servo power supply together.
5. On each servo plug, verify signal, positive, and ground against the markings on the Maestro before applying power. Do not rely only on wire color.

Connect the Maestro serial control wires

1. Locate the **Serial2** header labeled `G V R T` on the AstroPixels board.
2. Connect **T** on Serial2 to the serial receive input/RX on Maestro A.
3. Connect the same **T** serial line to the serial receive input/RX on Maestro B.
4. Connect **G** on Serial2 to ground on both Maestro controllers. Leave **V** and **R** disconnected.
5. In Pololu Maestro Control Center, set both controllers to **115200 baud**.
6. Give the controllers different device numbers. DroidLink_AP uses device number **12 for Maestro A** and **13 for Maestro B** by default.

Configure both boards in Maestro Control Center

Configure the two Maestros one at a time by USB before connecting them to DroidLink_AP. Do not assign both boards the same device number.

Maestro A

1. Connect only Maestro A to the computer by USB and open **Pololu Maestro Control Center**.
2. Open **Serial Settings**.
3. Set the serial mode to **UART, fixed baud rate** and set the baud rate to **115200**.
4. Set **Device Number** to **12**. Leave **Enable CRC** turned off; DroidLink_AP commands do not include a CRC byte.

5. Click **Apply Settings**. The number is not stored on the Maestro until the settings are applied.
6. Open **Channel Settings**. Set every channel used by a servo to **Servo**.
7. Set a safe **Min** and **Max** for each installed servo. These are the final pulse limits the Maestro will allow on that physical channel.
8. Click **Apply Settings** again, then disconnect Maestro A and label it **A - Device 12**.

Maestro B

Repeat the same procedure with Maestro B, but set **Device Number** to **13**. Click **Apply Settings** after changing Serial Settings and again after changing Channel Settings. Label this controller **B - Device 13**.

> **DroidLink endpoints do not replace the Maestro limits.** DroidLink_AP saves the requested Closed and Full pulse for each output, but the Maestro still enforces the Min and Max stored in Maestro Control Center. For example, if DroidLink_AP requests 2200 us while that channel's Maestro Max is 2000 us, the physical output stops at 2000 us. The same applies at the minimum end. A Maestro's Min and Max settings are the final controller-side safety boundary. Set them wide enough for the intended DroidLink calibration, but never wider than the servo and mechanism can safely tolerate.

Maestro A: dome panels

Maestro A replaces the first PCA9685 board shown as the Dome Panel Controller. Start at the Maestro's first physical output, which is printed as channel 0.

> **Why the number of outputs shown may not match the Maestro:** DroidLink_AP always reserves 18 output positions for Maestro A and another 18 for Maestro B. It does not detect whether the connected Maestro has 12, 18, or 24 physical channels. With a 12-channel Maestro, Web Config therefore shows six additional positions that do not physically exist. With a 24-channel Maestro, Web Config uses only physical channels 0 through 17, so the board's final six physical channels are not available to DroidLink_AP. This does not shift any assignments: Maestro B always begins with DroidLink_AP output 18, and its physical channel 0 remains the first holoprojector connection.

Dome connection	Maestro A physical channel	DroidLink_AP output
P1 / PANEL01	0 (first channel)	0
P2 / PANEL02	1	1
P3 / PANEL03	2	2
P4 / PANEL04	3	3
P5 / PANEL05	4	4
P6 / PANEL06	5	5
P7 / PIE PANEL 4	6	6
P8 / PIE PANEL 3	7	7
P9 / PIE PANEL 2	8	8
P10 / PIE PANEL 1	9	9
P11 / MINI PANEL 2	10	10
P12 / FRONT PSI PANEL	11	11
P13 / TOP PIE PANEL	12	12

An 18-channel Maestro is recommended for Maestro A because the standard layout uses 13 panel outputs. A 12-channel Maestro only has physical channels 0 through 11, so it cannot also hold P13. Do not move P13 to another channel unless its assignment is also changed in Web Config.

When using two 12-channel Maestros, P13 may be connected to Maestro B physical channel 6 and assigned to DroidLink_AP output 24. The six holoprojector servos remain on Maestro B physical channels 0 through 5, DroidLink_AP outputs 18 through 23. Adding P13 on channel 6 does not renumber or shift any holoprojector.

Maestro B: holoprojectors

Maestro B replaces the second PCA9685 board shown as the Holoprojector Controller. Again, begin with physical channel 0. DroidLink_AP shows Maestro B as outputs 18 through 35, so physical channel 0 on Maestro B appears as output 18 in Web Config.

Holoprojector connection	Maestro B physical channel	DroidLink_AP output
Front HP 1 / FRONT HOLO H	0 (first channel)	18
Front HP 2 / FRONT HOLO V	1	19
Rear HP 1 / REAR HOLO H	2	20
Rear HP 2 / REAR HOLO V	3	21
Top HP 1 / TOP HOLO H	4	22
Top HP 2 / TOP HOLO V	5	23

The remaining Maestro channels are optional and can be assigned in Web Config. Do not confuse a physical channel number on Maestro B with the DroidLink_AP output number. For example, **Maestro B physical channel 0 is DroidLink_AP output 18**.

Confirm the wiring before attaching linkages

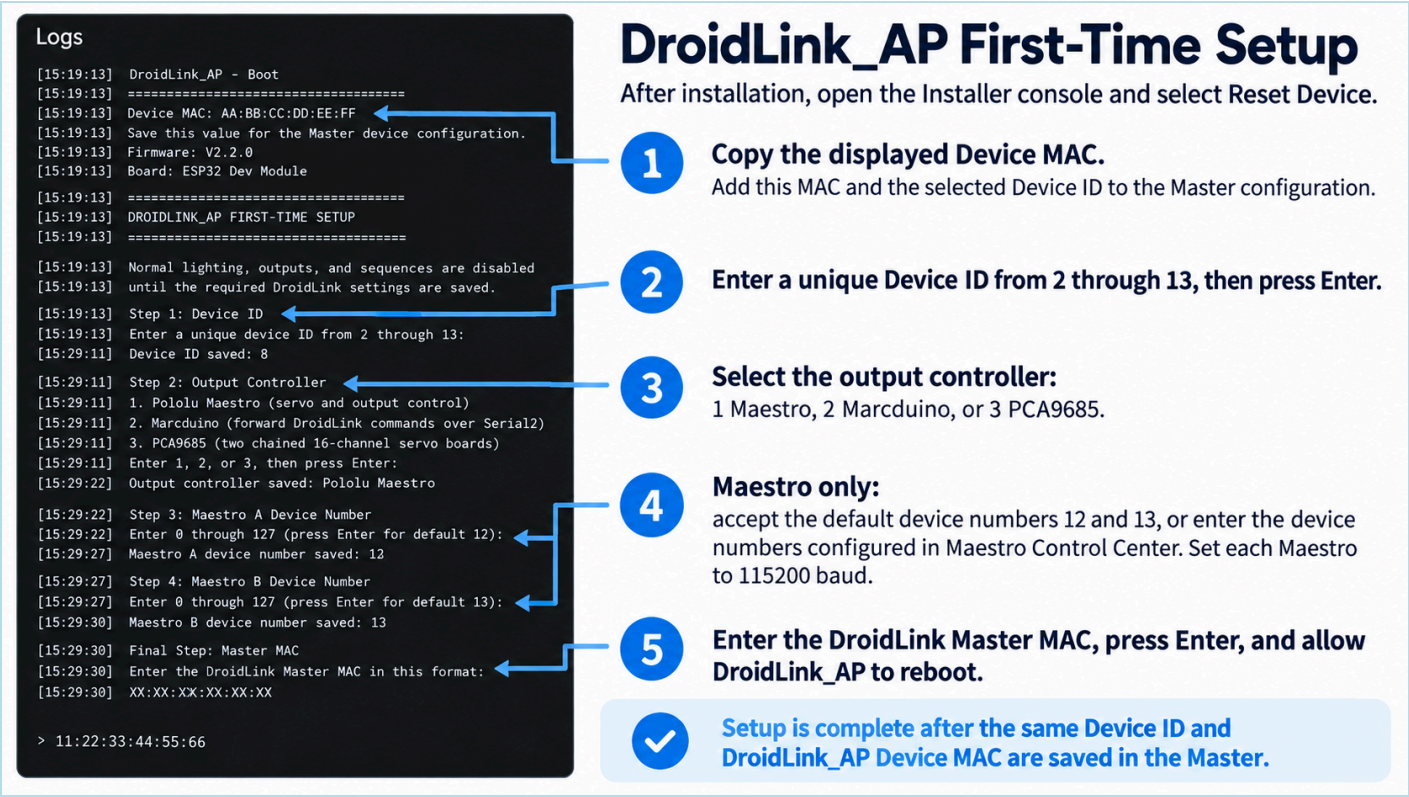
1. Install DroidLink_AP and complete the first-time setup below, selecting **Pololu Maestro** as the output controller.
2. Open Web Config and import the **Maestro dome template** linked later in this guide.
3. Compare every imported assignment with the two tables above before enabling live output.
4. Test one disconnected linkage at a time. Confirm that its Web Config name moves the expected physical servo.
5. Calibrate safe Closed and Full positions slowly, then save that output.
6. Reconnect the linkage only after both endpoints have been confirmed safe.
7. When all outputs work correctly, save a **Complete controller backup** from the Welcome page.

Install the firmware

If this is an update to an existing DroidLink_AP installation, first download a **Complete controller backup** from the Welcome tab. Install the update without selecting **Erase Flash** so the existing pairing and configuration can be retained. Use **Erase Flash** only for a new installation or when intentionally returning the device to a completely clean setup.

1. Open the DroidLink Installer in a supported browser.
2. Select **DroidLink_AP** and enter the DroidLink license key.
3. Connect the ESP32 by USB. For a new or intentionally clean installation, select **Erase Flash**. For an update that should retain its configuration, leave **Erase Flash** unselected.
4. Install the firmware and wait for installation to finish.
5. Open **Console Log** and reset the ESP32.
6. On a clean installation, follow the displayed first-time setup. Enter every requested value and press **Enter** after each response. An update with a valid retained configuration should restart without asking these questions.
7. For a clean installation, select the installed controller type, enter a unique DroidLink Device ID from 2 through 13, and enter the DroidLink Master MAC when requested. When selecting Maestro, accept device numbers 12 and 13 only if they match the numbers applied in Maestro Control Center.
8. Record the DroidLink_AP Device MAC shown in the console.
9. After DroidLink_AP restarts, open **Master System Setup**.
10. Select **Add Slave**, enter a useful name such as `DroidLink_AP`, and enter the exact Device MAC recorded from the console.
11. Save the Master configuration and allow the Master to restart if requested.

The console walkthrough below shows the information requested during first-time setup. The MAC addresses shown are examples; use the addresses displayed by your own devices.



DroidLink_AP first-time setup walkthrough

Open Web Config from the Watch Display

Before connecting to the `DL_AP` Wi-Fi network, download the correct template file to the phone, tablet, or computer that will be used for setup:

- Download the Maestro dome template
- Download the PCA9685 dome template

Use the **Maestro dome template** when the servos are connected to Pololu Maestro controllers. Use the **PCA9685 dome template** only when the servos are connected to PCA9685 boards. These files are not interchangeable because the second controller begins at a different DroidLink_AP output number.

Use the Watch Display to select DroidLink_AP by the Device ID assigned during first-time setup:

1. Power on the Master, Watch Display, and DroidLink_AP.
2. On the Watch Display, open **Command Center** and select **Settings**.
3. Scroll to the **DEVICE ID** buttons.
4. Select the Device ID assigned to DroidLink_AP.
5. Wait for DroidLink_AP to restart in Web Config mode.
6. On the phone, tablet, or computer containing the downloaded template, connect to the `DL_AP` Wi-Fi network.
7. Enter the default Wi-Fi password `droidlink`.
8. Open `http://192.168.4.1` in a web browser.

If the phone, tablet, or computer reports that this Wi-Fi network has no internet connection, remain connected. The network is only used to configure DroidLink_AP.

If selecting the Device ID does not open Web Config, confirm that the Watch Display is connected to the Master, the exact DroidLink_AP Device MAC is saved in Master System Setup, and the selected Device ID matches the ID assigned to DroidLink_AP during first-time setup. Then select the matching **DEVICE ID** again and wait for DroidLink_AP to restart.

Required: install the dome output template

Wiring the servos does not tell DroidLink_AP which panel or holoprojector is on each output. After Web Config opens, install the template before attempting to run panel commands or sequences.

Follow these steps exactly:

1. Select the **Outputs** tab at the top of Web Config.
2. Scroll down to **Servo/output templates**.
3. Select **Servo/output templates** to expand that section.
4. Click **Import template**.
5. The phone, tablet, or computer opens its file picker. Select `DroidLink-AstroPixels-Maestro-Dome-Template.json` when using Maestro controllers. Select `DroidLink-AstroPixels-PCA-Dome-Template.json` only when using PCA9685 boards.
6. Web Config displays **Assign template names to this device** followed by a long list of dropdowns. **These dropdowns are for advanced users creating a custom physical layout. Standard installations should ignore the entire dropdown list and must not change any of its selections.** The supplied template has already selected the correct standard assignments.
7. **Scroll all the way past the dropdown list to the bottom of the page.** The long list pushes the confirmation button far below where the template was imported.
8. Click **Apply template** at the bottom of the list. Wait for the status message confirming that the template and servo groups were applied.

The only reason to change a dropdown is when the physical wiring intentionally differs from the standard layout. For example, an advanced user with two 12-channel Maestros may move TOP_PIE_PANEL from output 12 to Maestro B physical channel 6, which is DroidLink_AP output 24. All other users should leave the dropdowns alone.

1. After the template is applied, **scroll back to the top of the Outputs page**.
2. In the output selector, choose the first configured servo. With the standard template this is **PANEL01 on output 0**. For Maestro, it operates P1 on Maestro A physical channel 0. For PCA, it operates P1 on the first physical output of PCA board 1.
3. Click **Enable live output** and carefully test the selected servo with its linkage disconnected. Confirm that the expected physical servo moves.
4. Establish safe Closed and Full positions, then click **Save** for that output. Disable live output before moving to unrelated setup work.
5. Repeat the select, test, calibrate, and save process for every installed servo, one output at a time.
6. If an assigned servo is not physically installed, select it and mark **Do Not Use This Output**. Do not leave a nonexistent mechanism active.
7. Expand **Saved output table** and verify that the expected panel and holoprojector names appear on the correct outputs.
8. Expand **Servo groups** and confirm that the imported panel and holoprojector groups are present.
9. Return to the **Welcome** tab and download a **Complete controller backup** after every output has been tested, calibrated, and saved.

> **Do not skip the template import.** Without it, the physical servo may be connected correctly but DroidLink_AP will not have the standard output names, assignments, and servo groups expected by the supplied commands and sequences.

What the template changes

Ready-to-import layouts are available in the DroidLink_AP template folder:

- Maestro dome template
- PCA9685 dome template

The template sets up output names, output types, assignments, and servo groups. It does not set the Device ID, Master MAC, lighting configuration, saved sequences, or safe servo endpoints. Existing calibrated endpoints are preserved, but a new installation must still have every servo calibrated and saved individually.

After configuration, use **Complete controller backup and restore** on the Welcome page to save the complete working setup.

Use **Exit Web Config** only after the template is applied, every installed servo has been calibrated and saved, the servo groups have been checked, and a complete controller backup has been downloaded.

Built-in `:SE` panel movements

With the supplied dome template applied, built-in `:SE` commands can run their AstroPixels lighting and sound together with the expected panel movement. The saved `ALL_PANELS` group controls normal all-panel movements, and `DOME_DANCE` controls the dome-dance movements. Do not rename those groups.

If one of those groups is missing, the lighting and sound can still run, but the related panel movement will not. A servo marked **Do Not Use This Output** remains blocked. DroidLink_AP supports up to 16 saved servo groups.

Optional fifth and sixth pie panels

The standard template configures four pie panels. Six-pie users can add the other two without changing the standard assignments:

1. Configure output rows 13 and 14 as `PIE_PANEL_5` and `PIE_PANEL_6`. These are Maestro A physical channels 13 and 14, or PCA A outputs A13 and A14.
2. Calibrate, save, and enable both outputs.
3. Add both outputs to `PIE_PANELS`, `DOME_DANCE`, and `ALL_PANELS`.
4. Confirm that `DOME_DANCE` contains 12 servos and `ALL_PANELS` contains 15.

During the built-in dome-dance sequences, pie 5 follows pie 4 and pie 6 follows pie 1. Users with four pie panels should leave the optional outputs unused.

Building LED and scrolling-text sequences

Each saved sequence can contain up to 512 timed actions.

Open **LED Sequence Builder** to combine timed AstroPixels lighting and scrolling-text actions under one saved `:AP00` through `:AP31` command.

1. Click **Add Seq Action** for a normal lighting action, or click **Add Scrolling Text** for a matrix-text action.
2. Set the action's **Start time (ms)**.
3. For scrolling text, choose the matrix target and color. Selecting **Both front logic matrices (different text)** displays separate **Top message** and **Bottom message** fields. DroidLink_AP stores those two messages as one local action.
4. Add the remaining lighting or text actions. Use the drag handle on an action card to change its saved order. Dragging does not alter its start time; order is most important when multiple actions share the same start time.

5. Click **Run sequence** and verify the complete result.
6. Enter a sequence name, choose an available DroidLink command, and click **Save Sequence**.

The Master sends only the saved `:APxx` command. DroidLink_AP loads and runs the complete timed sequence locally instead of requiring the Master to send every lighting and text action separately.

Advanced settings only: changing assignments and sharing sequences

This section is for users who intentionally wire a mechanism differently from the supplied template. A standard installation should leave every assignment dropdown unchanged, scroll to the bottom, and click **Apply template**.

What the assignment dropdowns mean

Each row under **Assign template names to this device** represents one named function, such as PANEL01 or FRONT_HOLO_H. The dropdown beside it selects the physical controller output that will perform that function.

For Maestro installations:

- `A0` is Maestro A physical channel 0, the first channel on that board. It is DroidLink_AP output 0.
- `A1` is Maestro A physical channel 1 and DroidLink_AP output 1.
- `B0` is Maestro B physical channel 0, the first channel on the second board. It is DroidLink_AP output 18.
- `B1` is Maestro B physical channel 1 and DroidLink_AP output 19.

For PCA installations, the first physical output on the first PCA board is also DroidLink_AP output 0. Therefore, **Maestro A channel 0 and the first output on PCA board 1 are both DroidLink_AP output 0**. This matching output number—not whether one controller is a Maestro and the other is a PCA—is what allows the same panel command to work on both installations.

Changing a function to another output

Only change an assignment when the mechanism is physically connected somewhere other than the standard location.

1. Find the function in the imported assignment list.
2. Open its dropdown and select the output where that mechanism is physically connected.
3. Do not assign two template functions to the same output. Web Config will not apply a template containing duplicate targets.
4. Scroll to the bottom and click **Apply template**.
5. Open **Saved output table** and confirm the function name appears on the new output.
6. Select that output, test the physical mechanism, calibrate its safe endpoints, and click **Save**.
7. Check **Servo groups**. Groups imported with the template follow the function to its selected output.
8. Download a new **Complete controller backup** after verifying the change.

Changing a dropdown assigns the function name, output type, and imported group membership to the selected output. It does not move a servo wire, create safe endpoints, or rewrite commands inside sequences that were already created.

For example, with two 12-channel Maestros, TOP_PIE_PANEL can be physically connected to Maestro B channel 6. In the TOP_PIE_PANEL dropdown, select `B6`, which is DroidLink_AP output 24. After applying the template, calibrate and save output 24. An older sequence containing a command for output 12 will still target output 12; its command must be changed to output 24.

What determines whether shared sequences work

DroidLink_AP sequences target **output numbers**. Custom output names are for the user's convenience and do not change sequence targeting. Two users may name the same panel differently and still exchange a sequence if that mechanism is assigned to the same DroidLink_AP output number on both devices.

Before sharing a sequence, both users should verify the following:

1. Both installations use the supplied template appropriate for their controller type.
2. Every mechanism used by the sequence has the same DroidLink_AP output number on both devices.
3. Any custom assignment is duplicated on the receiving device or the sequence commands are edited to match the receiving device.
4. Each user calibrates their own servo endpoints. A shared sequence does not transfer another user's physical calibration.

With the supplied standard templates, panel outputs P1 through P13 use DroidLink_AP outputs 0 through 12 in both Maestro and PCA modes. Panel sequences can therefore be exchanged without renaming the panels. AstroPixels lighting steps are also independent of Maestro versus PCA servo hardware.

Known V2.2.0–V2.3.0 holoprojector limitation

The current templates do not use matching DroidLink_AP output numbers for the holoprojectors:

Holoprojector function	Maestro template	PCA template
Front Holo H	18	16
Front Holo V	19	17
Rear Holo H	20	18
Rear Holo V	21	19
Top Holo H	22	20
Top Holo V	23	21

Consequently, panel and lighting steps remain correct when a V2.2.0 or V2.3.0 sequence is shared between Maestro and PCA users, but holoprojector steps might move a different holoprojector or do nothing. They will not be redirected to a dome panel when both users have applied the supplied standard templates.

For V2.2.0 and V2.3.0, users can correct the exported sequence JSON before importing it:

- Maestro sequence to PCA: change holo targets 18-23 to 16-21 respectively.
- PCA sequence to Maestro: change holo targets 16-21 to 18-23 respectively.

Only holoprojector command targets should be changed. Do not change panel targets 0 through 12 or AstroPixels lighting commands. Edit each complete command carefully so one replacement is not accidentally processed twice.

This holoprojector numbering mismatch is a known V2.2.0–V2.3.0 limitation and is planned to be corrected in a future DroidLink_AP release. Until that correction is actually released, verify or convert holo command targets before exchanging sequences between Maestro and PCA installations.

Backups and firmware updates

Download a **Complete controller backup** from the Welcome tab after the template, outputs, calibration, lighting, and sequences have been verified.

For a normal firmware update, use the DroidLink Installer with **Erase Flash** disabled so saved identity and configuration can be retained. Use **Erase Flash** only for a new installation, deliberate factory reset, or recovery that requires first-time setup again. Restore a backup only to a compatible controller setup.

Reset first-time setup

1. Connect the device to the DroidLink Installer by USB.
2. Open **Logs & Console**.
3. Type `NEWMAC` and press Enter.
4. Follow the first-time setup instructions.

Your saved device configuration and servo settings will not be erased.

Troubleshooting checklist

- Confirm the Device ID is unique from `2` through `13`.
- Confirm the exact DroidLink_AP Device MAC is saved with **Add Slave** in Master System Setup.
- Confirm the saved Master MAC is correct.
- Select the matching Watch Display **DEVICE ID** and wait for `DL_AP` to appear.
- Connect with password `droidlink` and open `http://192.168.4.1`.
- Confirm the imported template matches Maestro or PCA9685 hardware.
- Keep linkages disconnected until every installed servo has safe saved endpoints.

Completion checklist

- [ ] Correct controller type and unique Device ID saved
- [ ] Master MAC saved
- [ ] Exact Device MAC added to Master System Setup
- [ ] Web Config opens from the matching Watch Display **DEVICE ID** button
- [ ] Correct dome template applied
- [ ] Every installed servo calibrated and tested safely
- [ ] Lighting and saved sequences verified
- [ ] Complete controller backup downloaded